# ONA DUBEY

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LinkedIn GitHub Portfolio

## EDUCATION

**BTech in Data Science, Economics, and Business** (Plaksha University) 7.3
**Grade 12** (Shiv Nadar School, Noida) 82.4%
**Grade 10** (Kothari International School, Noida) 91%

## EXPERIENCE

**Engineering Intern | Airtel - XTelify** Gurugram, India, 2026

- Built **FlowCraft**, a **no-code chatbot builder on a React/FastAPI/MongoDB stack**, to help non-technical teams design, configure, and test multi-intent conversational bots without engineering involvement.

**Intern | PwC (PricewaterhouseCoopers) - Strategy&** Gurugram, India, 2025

- Conducted **competitive landscape and customer sentiment analysis** for Strategy&'s clients, focusing on **expansion strategies and market entry opportunities**.
- Synthesised research findings into strategic recommendations on product positioning and AI driven capability development.

**Technology Intern (Data Science) | OYO Rooms** Gurugram, India, 2024

- Derived actionable insights for OYO by analysing the **cost of losing a customer**, using measurable factors like **booking cancellations, public sentiment, repeat booking trends, and app activity**.
- Used **SQL through Metabase** for data extraction, transformation, and graphical analysis. I also studied public sentiment through online reviews and retention to quantify customer loss impact.
- Analysed app usage data, booking trends, stuck rates, and customer retention metrics across geographies and property types, leading to insights on rebooking behaviour and business optimisation.

**AI Project | EY (Ernst & Young)** Remote, India, 2024

- Predicted state-wise violent crime counts (2016-2021) across Indian states to aid law enforcement resource allocation.
- **Random Forest Regression and XGBoost** for predictive modeling, RMSE to assess accuracy. Basic data pre-processing and missing value imputation.
- Processed government crime data, graphically analysing high/low RMSE cases to refine forecasting and validate real-world applicability.

## PROJECTS

**Brain Injury Classification Using CT Scans** January-May, 2024

- Created an **ensemble of modified ResNet18 and GoogleNet CNNs**, trained on the CQ500 dataset's CT scans to classify up to 6 brain injuries.
- Pipeline: preprocessing, **region-growing segmentation**, the ensemble, DWSCSO optimiser, and final classification through PRSCNN.
- Multi-class, multi-label classification problem.
- **Accuracy on test set: 99.74%, Average F1 Score: 0.9986**

**Vehicle Signature Detection Using Surveillance Audio** January-April, 2025

- Built a **CNN-based multi-label classification pipeline** to identify 4 vehicle types from audio when visual surveillance fails.
- 4-layer CNN on de-noised, filtered **Mel/STFT spectrograms** (as 2D feature maps); Kaggle + Plaksha datasets.
- Kaggle: **99.6% accuracy vs 92.7% for fine-tuned YAMNet**; on noisier campus IoT-lab audio, YAMNet led (72.7%).

**NYC Taxi Demand Forecasting & Dynamic Pricing** January-April, 2025

- Predicted hourly taxi fare rates in Manhattan using **XGBoost regression** on 3M+ trip records to enable demand based (reverse) surge pricing.
- BSTS seasonal decomposition, engineered temporal features like cyclical encodings, multiple lag variables etc. ensuring chronological order.
- **$R^2$ = 0.9998, MAPE = 1.39%** for hourly fare prediction.

**Geopolitical Stress & Financial Risk: Evidence from India**

- Built **India-specific GPR / GPT / GPA geopolitical risk indices** from ~84,000 headlines (2010-2023) across 8 Indian outlets via LSEG/Refinitiv, using a 7-step proximity-search classification methodology.
- Regressed the indices against the **ADB Financial Stress Index via OLS**, full sample and a locally-filtered subsample.
- **GPT p = 0.0396, GPR (local) p = 0.0062**

## SKILLS

**Programming Languages:** MySQL, SQLite, Python
**Python Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch
**Domains:** Machine Learning, Deep Learning, Computer Vision, Data Analysis, Data Visualisation
**Additional Softwares:** Figma, Metabase